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Research Interests

HCI; XR/AR/VR; Human-AI interaction; Multimodal AI Interaction; Embodied Cognition; Serious Game; Child-Computer Interaction; Immersive Experience.

Education

KTH Royal Institute of Technology <i>MSc. Interactive Media Technology (GPA 4.26/5)</i>	Stockholm, SE Sep 2024 – Jun 2026 (Expected)
Huazhong University of Science and Technology <i>BSc. Electronic Information Engineering (GPA 3.39/4)</i>	Wuhan, CN Sep 2020 – Jun 2024

Research Positions

Hong Kong University of Science and Technology, Guangzhou <i>Research Assistant (Intern), Advised by Dr. Tengjia Zuo</i>	Guangzhou, CN Jul 2025 – Sep 2025
Led research on MR (mixed reality)-AI serious games, conducted user studies based on embodied learning principles, developed MR game environments on Meta Quest using Unity, and conducted quantitative and qualitative evaluations.	
Imperial College London, Royal College of Art <i>Visiting Student</i>	London, UK Jan 2022 – Feb 2022
- Conducted prototype design for an age-friendly wearable device. - Learned the Design Thinking methodology and applied it to conduct qualitative user research. - Designed service workflows and created storyboards.	

Publications

- Zhu, Y., Guo, L., **Sun, J.**, & Hei, X. (2024). Designing an LLM-driven Avatar System to Enhance Social Skills for Autistic Children in DTT Learning. In *2024 International Conference on Intelligent Education and Intelligent Research (IEIR)* (pp. 1–8). <https://doi.org/10.1109/ieir62538.2024.10959902>
- [Poster] Jiashu Sun. (2025) What if I actually Run in a Running game: A novel Seated Walking-in-Place Locomotion Method in Immersive VR Running Games. In The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)
- [submitted] Chen, D.*, Sun, J.*, Liu, S., & Zuo, T. (2026). Elementor: an Embodied Chemistry Learning Game Using Mixed Reality and Generative Artificial Intelligence. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*.

Honors

(National level, top 15%) Hubei Province Award, University Student Innovation & Entrepreneurship Training Plan Program,
 (National level, top 15%) Outstanding Member, University Student Innovation & Entrepreneurship Development Program,
 (University level, top 10%) Best Insight Award, University Maker Camp, Huazhong University of Science and Technology,

Research Projects

Elementor: an Embodied Chemistry Learning Game Using Mixed Reality and Generative Artificial Intelligence Supervised by Dr. Tengjia Zuo at HKUST-GZ	Jul 2025 – Sep 2025
- Developed an AI-powered MR serious game integrating chemistry learning with fantasy scenarios, which employs the LoRA model for anthropomorphic designs and LLMs for context-aware dialogues. Contributed novel AI-human codesign workflows and anthropomorphic storytelling approaches. - Conducted an exploratory study using quantitative questionnaires and qualitative interviews, giving design insights for MR serious games.	

What if I actually Run in a Running game: Evaluating a novel Seated Walking-in-Place Locomotion Method in Immersive VR Running Games

Supervised by Dr. Christopher Peters at KTH

Nov 2024 – Jan 2025

- Presented research on whether and how a new kind of WIP (Walking-in-Place) locomotion method, Cybershoes, could provide a more immersive VR running games experience by introducing foot movements as an additional interaction dimension.
- Conducted an experiment comparing Cybershoes with hand controllers using quantitative and qualitative methods.

Designing an LLM-driven Avatar System to Enhance Social Skills for Autistic Children

Supervised by Dr. Xiaojun Hei at HUST

Dec 2023 – Apr 2024

- Developed an interactive avatar system in Unity for autistic children, integrating Discrete Trial Training (DTT) methodology with Large Language Models (LLMs) to create an immersive environment for social skills learning.
- Engineered an AI-powered virtual therapy trainer using LLMs and voice synthesis to deliver personalized guidance and natural speech interactions.
- Implemented real-time Deep Learning models to analyze facial expressions and body movements, providing feedback to support children's social development.

Intelligent Traffic Surveillance System Based on Computer Vision

Supervised by Dr. Qiong Liu at HUST

Sep 2023 – Dec 2023

- Designed a system that dynamically adjusted traffic lights based on vehicle flow calculations and recorded traffic violations, providing support for traffic management and alleviating congestion during peak hours.
- Utilized the YOLOv5 deep learning algorithm (PyTorch framework) to detect vehicles in traffic scenes accurately. Utilized OpenCV and HyperLPR to identify vehicles' license plate information, enabling the detection and recording of violations.

Material Handling Robot Based on Visual Recognition Technology

Supervised by Dr. Lin Chen at HUST

Mar 2023 – Jun 2023

- Utilized Arduino boards for development to perform tasks such as grasping, transporting, and placing specific objects.
- Utilized deep learning models to achieve real-time image processing.
- Programmed robotic arm motion groups and collaborative work with the infrared sensor tracking module.

“Embracing”: design a device for the elderly’s healthcare during the COVID-19 pandemic

Supervised by Dr. Nick de Leon, RCA

Jan 2022 – Feb 2022

- Applied the Double-Diamond model to design a hi-fi model, aiming at providing a guide for the elderly when going to the hospital.
- Conducted interviews to gather user insights, created storyboards to visualize design concepts, and designed and developed a prototype of a wearable device.

Skills

Programming & Data Science: Python, C, C#, Machine learning, Deep learning, SPSS, HTML

Hardware Devices: Arduino, Oculus, Furhat robotics, FPGA

Design Tools: Unity, Figma, PS, Procreate, Blender, C4D

Research Methods: Interview, Case study, Thematic Analysis, User Journey, Perception Measurement

Languages: English – CEFR C1 (IELTS 7), Chinese – Native